

EE 3TR4 (COMMUNICATION SYSTEMS): QUIZ 2

INSTRUCTOR: DR. JUN CHEN

Student Name: _____

Student Number: _____

A white noise process $W(t)$ with power spectral density $S_W(f) = 0.004 \text{ mW/Hz}$ is added to the signal $x(t) = \cos(100\pi t) + 2\sin(2000\pi t)$. The resulting process is passed through an ideal low-pass filter with transfer function $H(f) = \text{rect}(\frac{f}{1000})$. Determine the signal-to-noise ratio at the filter output.

Fourier transform of special functions:

$$\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(f),$$

$$\text{triang}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}^2(f),$$

where

$$\text{rect}(t) = \begin{cases} 1, & |t| < \frac{1}{2}, \\ 0, & \text{otherwise}, \end{cases}$$

$$\text{triang}(t) = \begin{cases} 1 - |t|, & |t| < 1, \\ 0, & \text{otherwise}, \end{cases}$$

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}.$$

Properties of Fourier transform:

If $g(t) \xleftrightarrow{\mathcal{F}} G(f)$ and $h(t) \xleftrightarrow{\mathcal{F}} H(f)$, then

$$g(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} G\left(\frac{f}{a}\right),$$

$$G(t) \xleftrightarrow{\mathcal{F}} g(-f),$$

$$g(t - t_0) \xleftrightarrow{\mathcal{F}} G(f) e^{-j2\pi f t_0},$$

$$g(t) e^{j2\pi f_c t} \xleftrightarrow{\mathcal{F}} G(f - f_c),$$

$$g(t) \cos(2\pi f_c t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} [G(f - f_c) + G(f + f_c)],$$

$$g(t) * h(t) \xleftrightarrow{\mathcal{F}} G(f) H(f),$$

$$g(t) h(t) \xleftrightarrow{\mathcal{F}} G(f) * H(f).$$

Random process:

The autocorrelation function and power spectral density (PSD) form a Fourier transform pair:

$$R_X(\tau) \xleftrightarrow{\mathcal{F}} S_X(f).$$

For a linear time-invariant filter with transfer function $H(f)$, the input PSD $S_X(f)$ and output PSD $S_Y(f)$ are related by

$$S_Y(f) = S_X(f) |H(f)|^2.$$

Solution to Quiz 2

At the filter output, the mean noise power is $(1000\text{Hz}) \times (0.004\text{mW/Hz}) = 4\text{mW}$ while the mean signal power is 0.5W since only the $\cos(100\pi t)$ component passes through the filter. Therefore, the resulting signal-to-noise ratio is $\frac{0.5\text{W}}{4\text{mW}} = 125 \approx 21\text{dB}$.

EE 3TR4 C02 (COMMUNICATION SYSTEMS): QUIZ 2

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A random process $X(t)$ with autocorrelation function

$$R_X(\tau) = \frac{A^2}{2} \cos(2\pi f_c \tau)$$

is input to a filter with frequency response

$$H(f) = \frac{1}{1 + j2\pi RCf}$$

(a) What is the autocorrelation function of the output process?

(b) What is the average power content of the output process?

a) $R_Y(\tau) = ?$

$$S_Y(f) = S_X(f) |H(f)|^2$$

$$= \frac{A^2}{4} [\delta(f-f_c) + \delta(f+f_c)] \cdot |H(f)|^2$$

$$R_Y(\tau) = \int_{-\infty}^{\infty} S_Y(f) e^{j2\pi f\tau} df$$

$$= \int_{-\infty}^{\infty} \frac{A^2}{4} [\delta(f-f_c) + \delta(f+f_c)] \cdot |H(f)|^2 \cdot e^{j2\pi f\tau} df$$

$$= \frac{A^2}{4} \int_{-\infty}^{\infty} \delta(f-f_c) \cdot |H(f)|^2 \cdot e^{j2\pi f\tau} + \int_{-\infty}^{\infty} \delta(f+f_c) \cdot |H(f)|^2 \cdot e^{j2\pi f\tau}$$

$$= \frac{A^2}{4} \left(|H(f_c)|^2 \cdot e^{j2\pi f_c \tau} + |H(-f_c)|^2 \cdot e^{-j2\pi f_c \tau} \right)$$

$$= \frac{A^2}{4} \cdot \frac{1}{1 + (2\pi RCf_c)^2} \left(e^{j2\pi f_c \tau} + e^{-j2\pi f_c \tau} \right)$$

$$= \frac{A^2}{4} \cdot \frac{1}{1 + (2\pi RCf_c)^2} \cdot 2 \cos(2\pi f_c \tau)$$

$$= \frac{A^2}{2} \cdot \frac{1}{1 + (2\pi RCf_c)^2} \cos(2\pi f_c \tau)$$

$$S_X(f) = \mathcal{F}\{R_X(\tau)\}$$

$$= \mathcal{F}\left\{\frac{A^2}{2} \cos(2\pi f_c \tau)\right\}$$

$$= \frac{A^2}{2} \cdot \frac{1}{2} [\delta(f-f_c) + \delta(f+f_c)]$$

$$= \frac{A^2}{4} [\text{~~~~~}]$$

b) $P_Y = R_Y(0)$

$$= \frac{A^2}{2} \cdot \frac{1}{1 + (2\pi RCf_c)^2} \cos(2\pi f_c \cdot 0)$$

$$= \frac{A^2}{2} \cdot \frac{1}{1 + (2\pi RCf_c)^2}$$

Fourier transform of special functions:

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$$S_Y(f) = S_X(f) |H(f)|^2.$$

Solution to Quiz 2

- (a) Autocorrelation function of the output process. [8 Marks]

Using the Fourier transform pair

$$\cos(2\pi f_c t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} [\delta(f - f_c) + \delta(f + f_c)],$$

the spectral density $S_X(f)$ is obtained as

$$S_X(f) = \frac{A^2}{4} [\delta(f - f_c) + \delta(f + f_c)].$$

Hence,

$$\begin{aligned} S_Y(f) &= S_X(f) |H(f)|^2 \\ &= \frac{A^2}{4} |H(f)|^2 [\delta(f - f_c) + \delta(f + f_c)]. \end{aligned}$$

Using the inverse Fourier transform of $S_Y(f)$ to evaluate $R_Y(\tau)$, we get

$$\begin{aligned} R_Y(\tau) &= \int_{-\infty}^{\infty} S_Y(f) e^{j2\pi f\tau} df \\ &= \int_{-\infty}^{\infty} \left\{ \frac{A^2}{4} |H(f)|^2 [\delta(f - f_c) + \delta(f + f_c)] \right\} e^{j2\pi f\tau} df \\ &= \frac{A^2}{4} \int_{-\infty}^{\infty} \{ |H(f)|^2 e^{j2\pi f\tau} \} [\delta(f - f_c) + \delta(f + f_c)] df \\ &= \frac{A^2}{4} \int_{-\infty}^{\infty} \{ |H(f)|^2 e^{j2\pi f\tau} \} \delta(f - f_c) df + \frac{A^2}{4} \int_{-\infty}^{\infty} \{ |H(f)|^2 e^{j2\pi f\tau} \} \delta(f + f_c) df \\ &= \frac{A^2}{4} (|H(f_c)|^2 e^{j2\pi f_c\tau}) + \frac{A^2}{4} (|H(-f_c)|^2 e^{-j2\pi f_c\tau}) \\ &= \frac{A^2}{4} \frac{1}{1 + (2\pi RC f_c)^2} (e^{j2\pi f_c\tau} + e^{-j2\pi f_c\tau}) \\ &= \frac{A^2}{4} \frac{1}{1 + (2\pi RC f_c)^2} (2 \cos(2\pi f_c\tau)) \\ &= \frac{A^2}{2} \frac{1}{1 + (2\pi RC f_c)^2} \cos(2\pi f_c\tau). \end{aligned}$$

Thus,

$$R_Y(\tau) = \frac{A^2}{2} \frac{1}{1 + (2\pi RC f_c)^2} \cos(2\pi f_c\tau).$$

- (b) Average power content of the output process. [2 Marks]

$$P_Y = \int_{-\infty}^{\infty} S_Y(f) df = R_Y(0) = \frac{A^2}{2} \frac{1}{1 + (2\pi RC f_c)^2}.$$